Optical Disks

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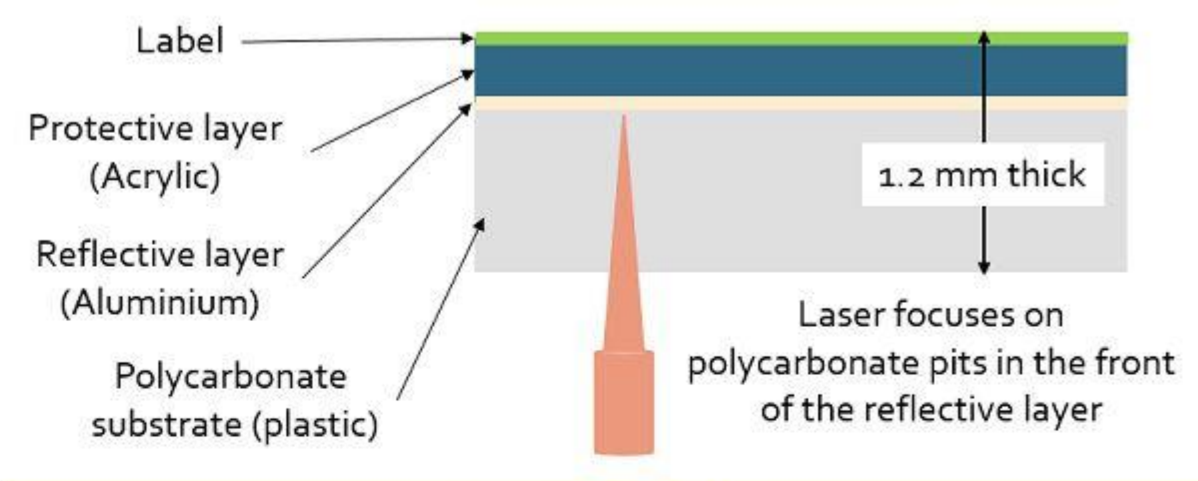
Optical Disks

- **Optical Disc** (**OD**) is a flat, usually circular disc that encodes binary data (bits) in the form of pits and lands.
- A computer memory that uses optical techniques which generally involve an
  - addressable laser beam
  - a storage medium which responds to the beam for writing and sometimes for erasing
  - a detector which reacts to the altered character of the medium when it uses the beam to read out stored data.

Compact Disk

- The **CD (Compact Disc)** was the first step towards the idea of digital encoding of the data.
- It uses a unique method of encoding in which a 14-bit code indicates a byte and this encoding technique also helps in error detection.
- The CDs first layer is constructed by the polycarbonate plastic which stores the data in the indented pits. This layer also provides a clear glass base. The flat or unindented area is known as the land.
- The thin aluminium material covers the programmed disk, and a protective acrylic then coats this aluminium layer.
- In the last part, the disk is stamped by a label. The pits are organised into the spiral track, in the outward direction from the middle of the disk, though decreasing the risk of damage by storing the index information at the start of the disk. , and the length and width of the pit are 0.8-3 and 0.5 microns respectively.

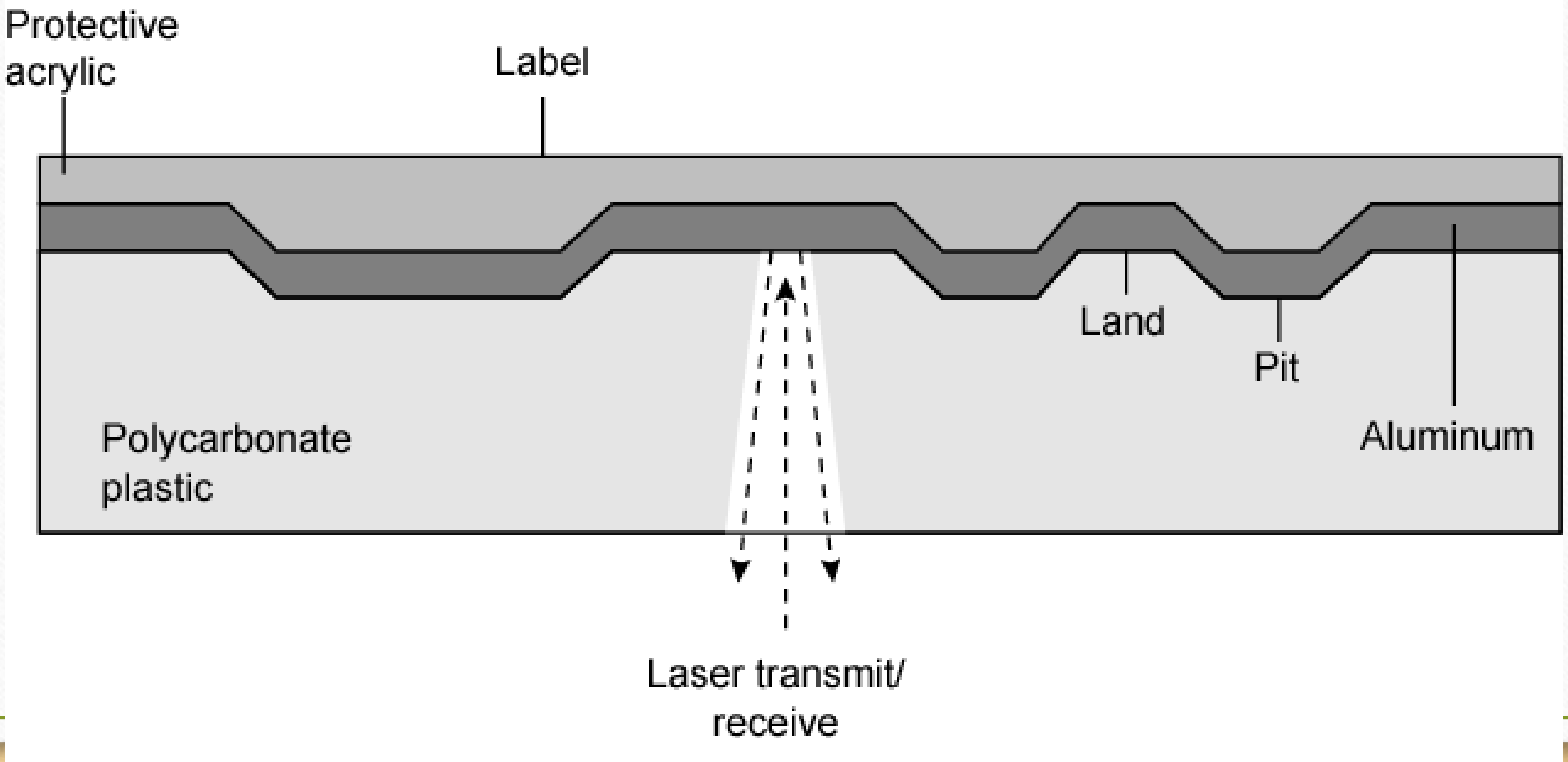
Layers of a CD



Layers of a CD

CD Operation

- The stored bits are read through targeting the laser light on to the polycarbonate part of the spinning disk. Indented portion of the disk reflects back the light which is detected by the photodetector. There are several variations of the compact disc such as CD-R, CD-ROM, CD-RW.



Pros and Cons of CD

Advantages

- The optical disk together with the information stored on it can be mass replicated inexpensively;-unlike a magnetic disk.
- The optical disk is removable.

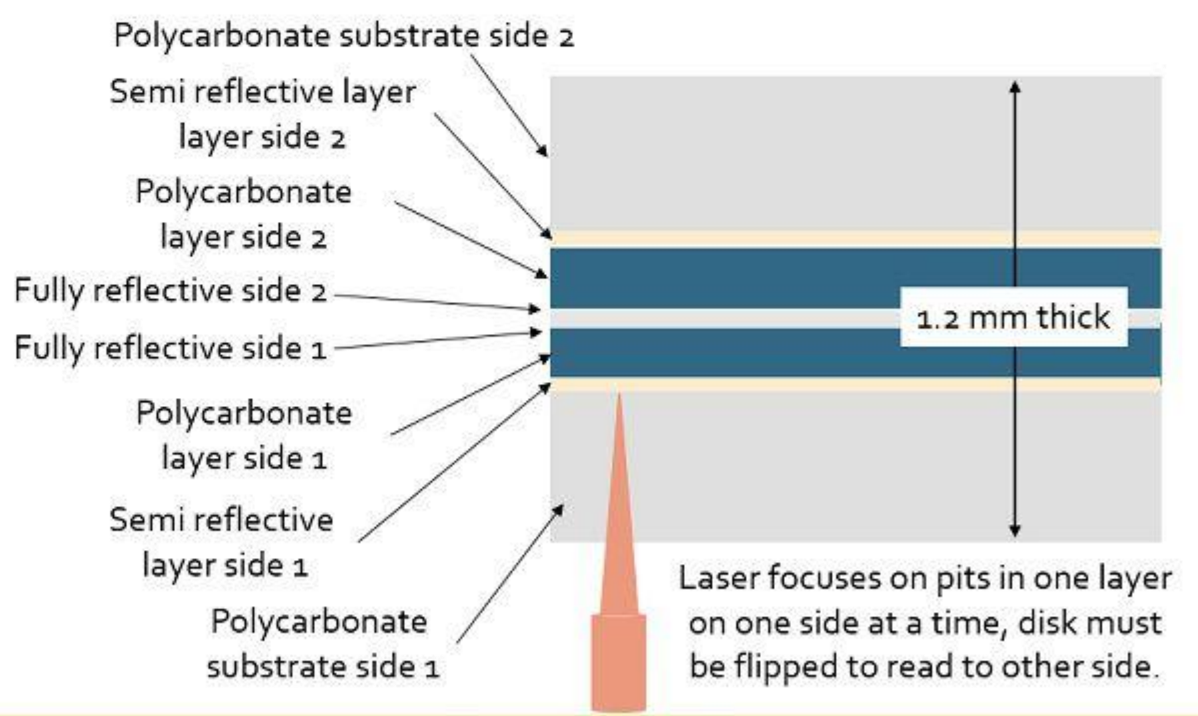
Disadvantages

- It is read only and cannot be updated.
- It has an access time much longer than that of a magnetic disk drive.

Digital Versatile Disk (DVD)

- The advent of **DVD (Digital Versatile Disk)** provided an alternative for the videotape used in VCR (Video Cassette Recorder) and CD-ROM used in PC as the DVD can acquire 7 times larger amount of the data relative to CD.
- It renders videos with excellent picture quality and random access. A DVD is built from the same material as the CD but the process and the layers are different, it is used from both of the sides such as two CDs are sticking together.
- The reasons why it provides greater size than a CD are – dense packing of the bits, use of shorter wavelength laser and generation of the two-sided disc (data can be read from write to both the sides). To obtain the smaller spacing about 0.74 microns between the spiral tracks and the minimum distance of 0.4 microns between the pits a short wavelength laser is used.

Layers of a DVD



Layers of a DVD

DVD Operation

- It also employs a dual layer of pits and lands on the standard layer and a semi-reflective layer on the top of the reflective layer. To read the content of the layers separately in a DVD the laser must be carefully focused. Similar to CD, DVD also has variants such as DVD-R, DVD-RW.

Key Differences Between CD and DVD

- A compact disk can hold up to 700 MB of data while a digital versatile disk can hold a maximum 17 GB of data.
- As the DVD can acquire a larger size, it is more prevalently used than a CD.
- The metal layer used for recording is placed beneath the labelling side in the CD. In contrast, in DVD this metal layer is situated in the center of the disk.
- The layers of pits and land on a CD can be only one whereas in DVD it can be two.
- A CD contains the spacing of 1.6 micrometers between spiral tracks and 0.834 micrometers between the pits on the disc. On the other hand, the spiral loops in DVD are 0.74 micrometers apart and the distance between the pits is 0.4 micrometers.
- The removal of the adhesive label of a CD can cause the severe damage to the CD. On the contrary, when the adhesive label of the DVD is removed, the imbalance in spin is caused.

Blu-ray disc

- **Blu-ray** is a high definition optical disk which can hold a greater storage space as compared to other optical storage devices. It was mainly devised to store high definition videos, and this was not possible in the previous DVD technology.
- The Blu-ray name is originated by merging blue and optical "ray" and this because it uses blue-violet laser instead of a red laser. As the blue laser's wavelength is shorter, it helps in achieving higher bit density. Due to the shorter wavelength, the data pits in the high definition optical disk are tiny. The data pits are used to form the digital 1's and 0's.
- Blu-ray can store the enormous amount of data as compared to other disk formats, and the reason behind this is the less gap between the data layer on the disk and the laser. It generates tighter focus, less distortion and smaller pits (and tracks) as an outcome of placing the laser closer to the disk. The maximum storage space of Blu-ray is 25 GB on a single side.
- The three versions of Blu-ray are
  - Read-only BD-ROM
  - Recordable once BD-R
  - Recordable BD-RE

Key Differences Between Blu-ray and DVD

- The blu-ray optical disks perform better than DVD in terms of storing high-quality data, comparatively.
- In DVD the red laser is used to fetch data whose wavelength is 650 nm while blu-ray uses a blue-violet laser in which wavelength is 405 nm.
- DVD can store 4.7 GB in a single layer and a maximum of 7.4 in the double layer. As against, blu-ray has maximum space of 25 GB in a single layer while 50 GB in a dual layer.
- Blu-ray provides a higher data rate that is up to 36 Mbps whereas the DVD rate of data transfer is 11.08 Mbps.
- Blu-ray is more secure and protected.

Advantages of Blu-ray and DVD

Advantages of Blu-ray

- It has a large amount of capacity.
- Compulsory managed copy.
- It is backward compatible.
- Quality support
- Durable HDTV support

Advantages of DVD

- Provides high density as compared to CD.
- Cost effective relative to CD.
- Compatible with most older and current DVD writers.
- Durable.

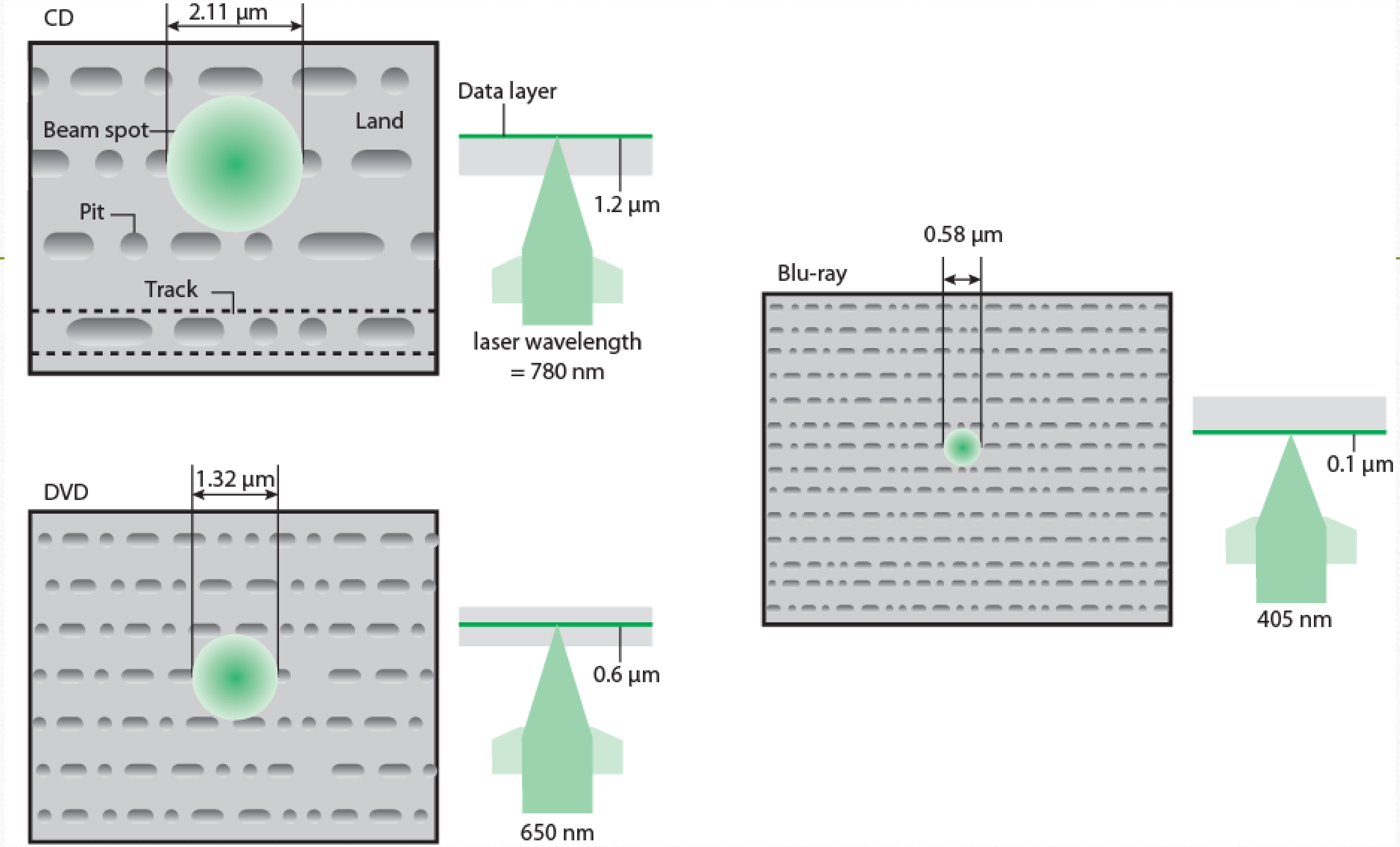
Disadvantages of Blu-ray and DVD

Disadvantages of Blu-ray

- Its cost is quite high.
- However, the blu-ray is capable of storing high definition videos but in a limited way.
- Competitors risk.

Disadvantages of DVD

- Data transfer is slow as compared to blu-ray.
- Storage capacity less and can not store HD videos.


CD
2.11 µm
Beam spot
Land
Pit
Track
Data layer
1.2 µm
laser wavelength
= 780 nm
DVD
1.32 µm
0.6 µm
650 nm
0.58 µm
Blu-ray
0.1 µm
405 nm